4.6. Material processing using longitudinally single-mode UV ns laser pulses

The energy dependence of cavitation bubble size presented in figures 5 and 6 confirms the possibility of creating fine adjustable nanoeffects by slm UV ns pulses, with stepwise transition to a more disruptive microregime. Figure 12 now demonstrates that the slm UV ns pulses can be utilized for micro-material processing such as corneal refractive surgery, micro-patterning inside glass, and the generation of refractive index modifications.

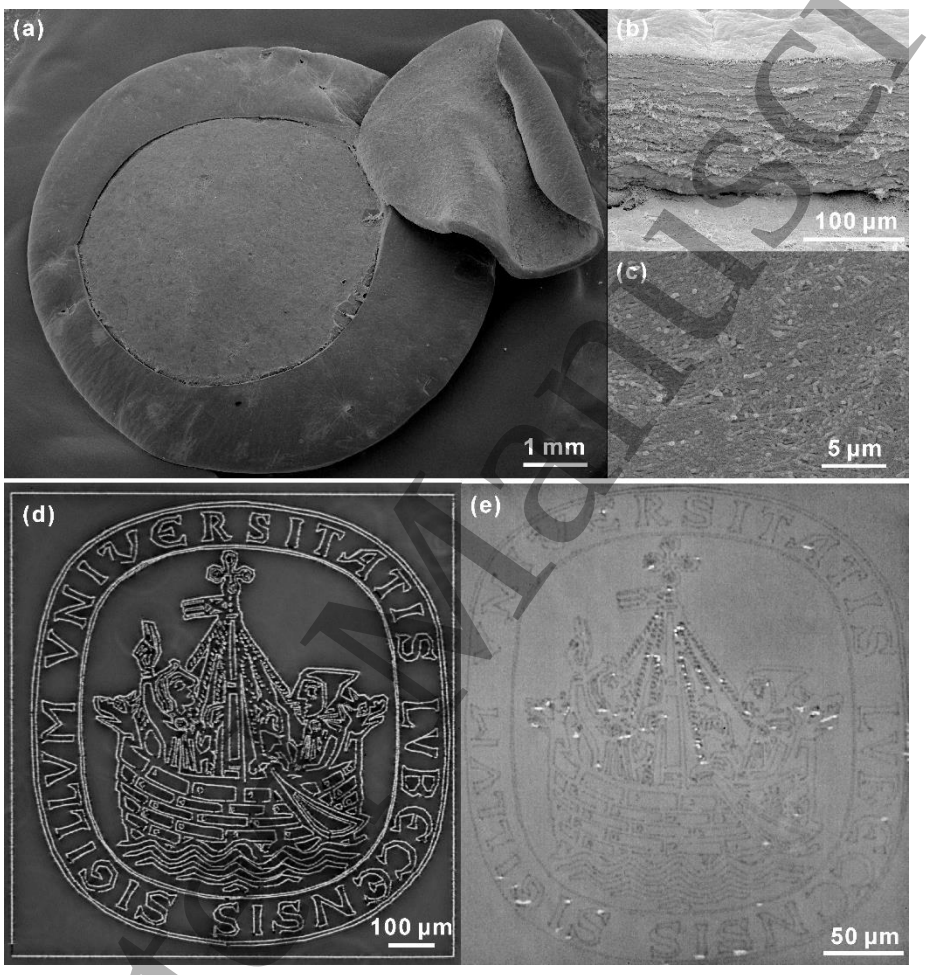

Figure 12 Micro-material processing by means of 0.56-ns, 355-nm single longitudinal mode laser pulses applied with 1 kHz pulse repetition rate. (a) Scanning electron micrograph of a 6-mm LASIK flap in porcine cornea produced with 6 µm separation between focal spots. (b) Side cut of the flap. Due to the high cutting precision, individual corneal lamellae can be identified. (c) Enlarged view of the flap bed. (d) Logo of the University of Luebeck written within a microscope glass slide (height 1 mm). (e) Phase contrast image of the logo written at lower pulse energy, height 300 µm. The laser pulses have produced small, light scattering cavities in (d) and refractive index changes in (e). Bright spots indicate glitches, where a void was created. Laser pulses were focused at $NA = 0.28$ in (a) and (c), and at $NA = 0.75$ in (b), (d), and (e). Laser pulse energies were 1.0 µJ in (a) and (c), 2 µJ in (b), 0.44 µJ in (d), and 0.27 µJ in (e).

Accepted Manuscript

In state-of-the-art "Femto-LASIK", a flap is cut into the anterior corneal stroma using fs laser pulses of 1030-1040 nm wavelength. The pulses are focused in a raster pattern into the desired dissection plane to produce the cut, and the flap is then lifted for subsequent excimer laser ablation [13,105,241]. Figs. 12(a) to (c) demonstrate that UV-A ns pulses can fulfill the same task as fs laser pulses. The cutting precision is even better because, at equal *NA*, focal diameter and length at 355 nm are only 1/3 of the values for IR Femto-LASIK [17].

Figs. 12 (d), and (e) show patterns in borosilicate glass produced at 1 kHz pulse repetition rate. At 0.44 µJ pulse energy, the expanding plasma creates tiny light-scattering cavities (Fig. 12d), while at an energy of 0.27 µJ mainly refractive index changes are produced (Fig. 12e). The size of individual laser effects as determined by observation with a phase contrast microscope is < 1µm, right at the optical resolution limit. The energy range in which refractive index changes can be formed without disrupting the glass corresponds to the small bubble regime in water (Figs. 5 and 6), while cavity formation occurs in the BPL regime. In this analogy, the transition from refractive index formation to cavity formation in glass resembles the abrupt increase of the effects size in water at the upper end of the small-bubble regime. The bright spots visible in Fig 12(e) are glitches arising from localized transitions into the BPL regime.

Accepted Manuscript

5. Discussion

5.1. Deterministic vs stochastic breakdown behavior

Our experimental investigations showed that breakdown thresholds are much sharper for femtosecond and single-longitudinal-mode nanosecond laser pulses with smooth temporal shape than for ns multimode pulses exhibiting intensity spikes from longitudinal mode beating [Figure 4(b) and Table 2]. Spikes in ns pulses facilitate multiphoton seed electron generation, which lowers the threshold [Fig 4(a)], and the pulse-to-pulse variations of spiking behavior introduce a stochastic component [Fig. 5(b)]. Similar correlations between laser mode and breakdown behavior have been reported previously on breakdown in bulk SiO_2 and silicate glasses [231,232]. At sample surfaces, threshold fluctuations were observed even with slm pulses [127,232] but these fluctuations were later found to be due to remnant impurities from the surface polish and were absent in pure SiO_2 [231]. A correlation between laser mode and breakdown behavior is also found in the $R_{max}(E_L)$ curves in Fig. 5. They are smooth for slm pulses but the bubble sizes exhibit strong statistical variations for multimode laser operation. Thus, with appropriate laser mode control also nanosecond breakdown in transparent dielectrics is a reproducible, "deterministic" event as long the impurity content is low.

Highly reproducible breakdown behavior in distilled water for tight focusing was observed at numerical apertures down to $NA = 0.3$ (Fig. 6) because under these conditions, hardly any impurities are present in the focal volume. The density of particulate impurities is $\approx 2.5\times10^4$ cm^{-3} in distilled and $\approx 1.2\times10^6$ cm^{-3} in tap water [201]. For $NA = 0.3$ and $\lambda = 355$ nm, one obtains a very small probability $P \approx 10^{-4}$ % of finding a particle within the focal volume in distilled water, and for tap water the probability is $P \approx 5\times10^{-3}$ %. It is still only about 9% for $NA = 0.1$ but it rises quickly, when the NA is further reduced, which explains the previously reported stochastic behavior for small focusing angles [109,200,201]. Thus, the conditions for reproducible breakdown behavior are reproducible laser pulse shape, low impurity density in the medium, and tight focusing.

For biological tissues, cells and cell culture media, stronger statistical threshold fluctuations have been observed [193,242,243]. This applies especially to IR ns breakdown, where the seed electrons for avalanche ionization are the critical hurdle for the onset of breakdown. Since the cross section for multiphoton ionization depends on the excitation energy, biomolecules can lower I_{th} if they possess intermediate energy levels between valence and conduction band of water that facilitate multiphoton-generation of seed electrons. In UV ns breakdown, where seed electrons are always readily available,

Accepted Manuscript